CHAPTER 02

Isomers

Independent exercise solving

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

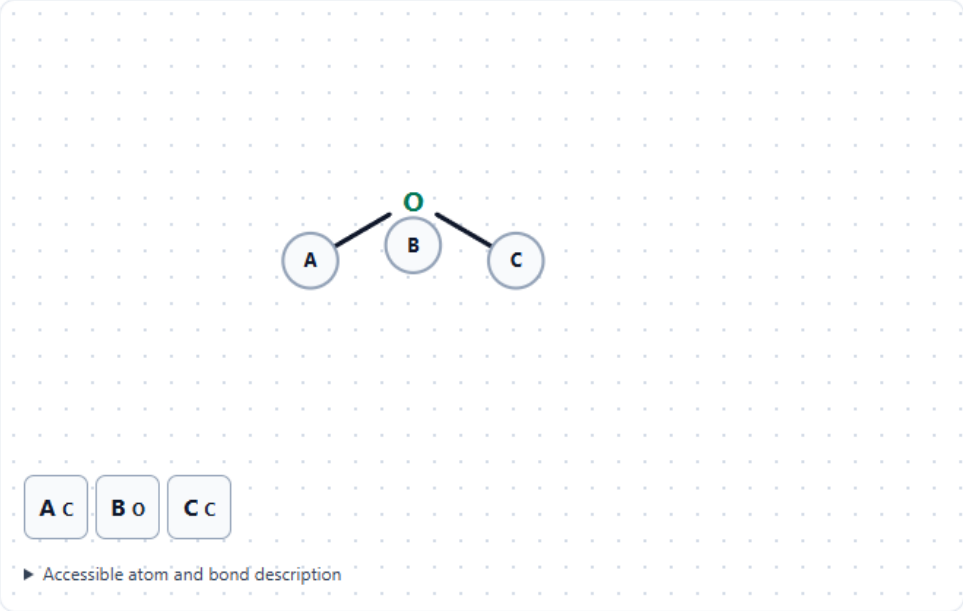
Source snapshot SHA-256: 3790c3072a422920a2793636cc0e671c8e2a766731209b66b68c3e110873ccb4

BLIND PASS: answer keys, author molecule names/formulas, hints, explanations and answer overlays are withheld. Solve every numbered stage independently. Report the stage ID, answer, derivation, and uncertainty. Question wording and displayed diagram captions are retained because they are visible task data.

Do not open the full review packet in this chat before saving your independent attempt. The full packet contains keys and lessons that would contaminate a blind attempt.

isomer-pair | Independent exercise

Screenshot: isomer-pair-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=60.00, y=110.00; charge=0; explicit H override=none

1 (B): O; x=120.00, y=75.00; charge=0; explicit H override=none

2 (C): C; x=180.00, y=110.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1]]

isomer-pair:stage-1 | Select the oxygen atom.

Answer type: atoms

isomer-pair:stage-2 | Compared with ethanol (C2H6O), this compound is a…

Answer type: choice

Option 0 (A): Constitutional isomer

Option 1 (B): Conformer

Option 2 (C): Enantiomer

isomer-pair:stage-3 | Which can donate an O–H hydrogen bond?

Answer type: choice

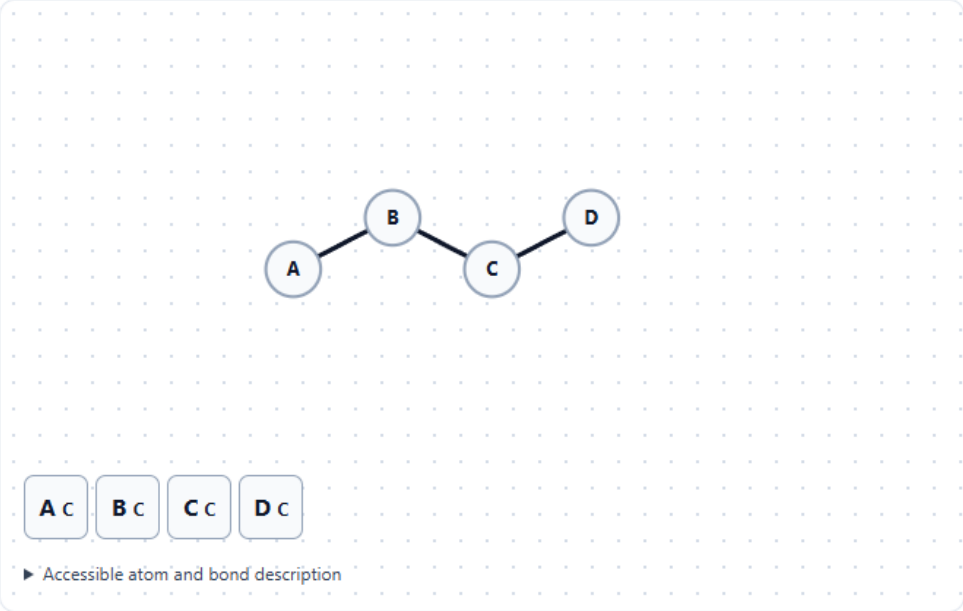
Option 0 (A): Both

Option 1 (B): Ethanol only

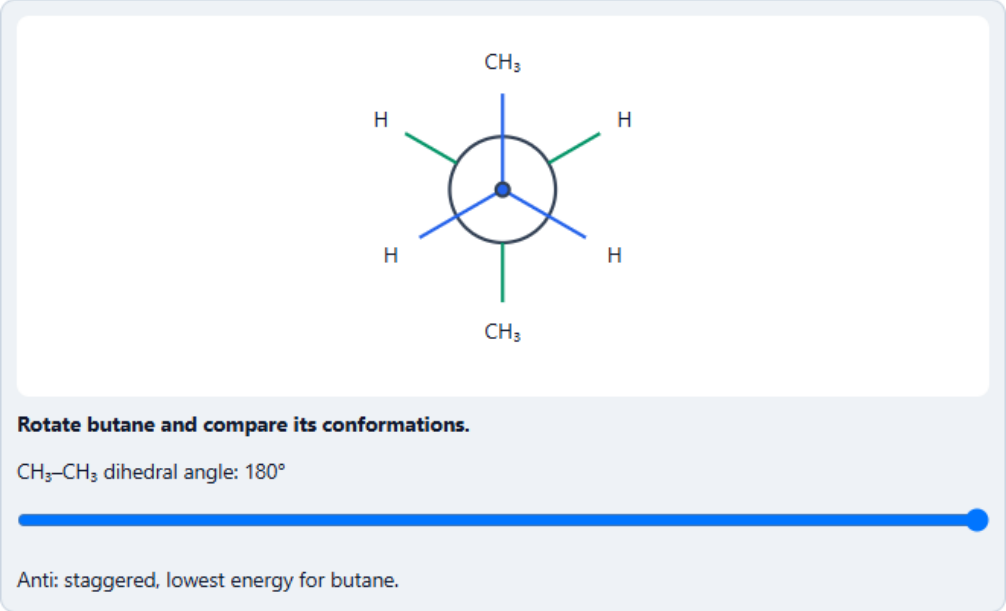
Option 2 (C): This ether only

butane-rotation | Independent exercise

Screenshot: butane-rotation-stage-1-molecule



Screenshot: butane-rotation-diagram



{"diagramSpecification": {"type": "newman", "caption": "Rotate butane and compare its conformations."}}

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1]]

butane-rotation:stage-1 | Select the two carbons forming the central bond.

Answer type: atoms

butane-rotation:stage-2 | At a 180° methyl–methyl dihedral, the conformation is…

Answer type: choice

Option 0 (A): Anti

Option 1 (B): Gauche

Option 2 (C): Fully eclipsed

butane-rotation:stage-3 | A cyclohexane chair flip changes…

Answer type: choice

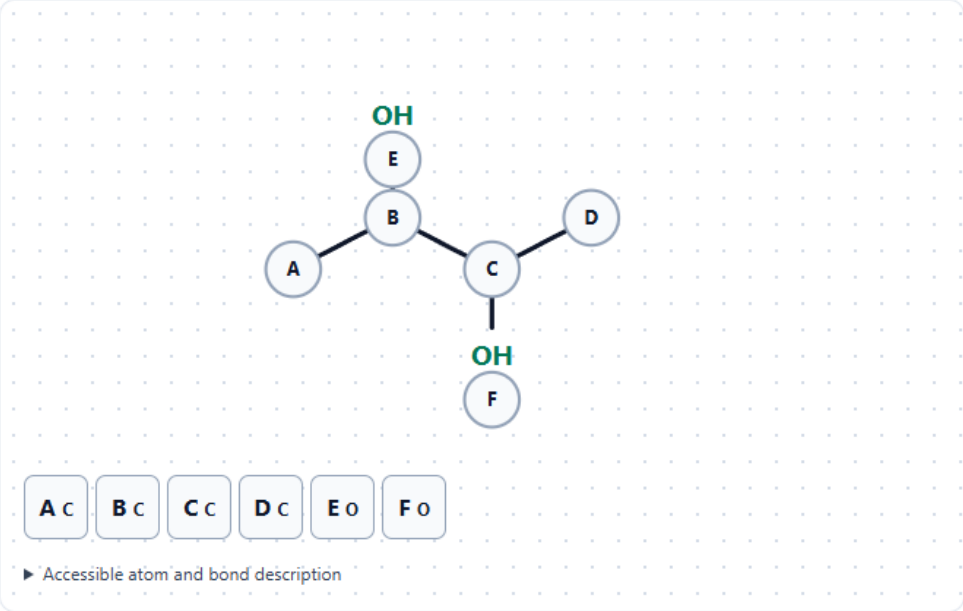
Option 0 (A): Up into down

Option 1 (B): Axial into equatorial, preserving up/down

Option 2 (C): Cis into trans

stereocenters | Independent exercise

Screenshot: stereocenters-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

4 (E): O; x=108.00, y=25.00; charge=0; explicit H override=none

5 (F): O; x=166.00, y=165.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [1, 4, 1], [2, 5, 1]]

stereocenters:stage-1 | Select both potentially stereogenic carbons.

Answer type: atoms

stereocenters:stage-2 | What is the maximum 2^n count before symmetry is considered?

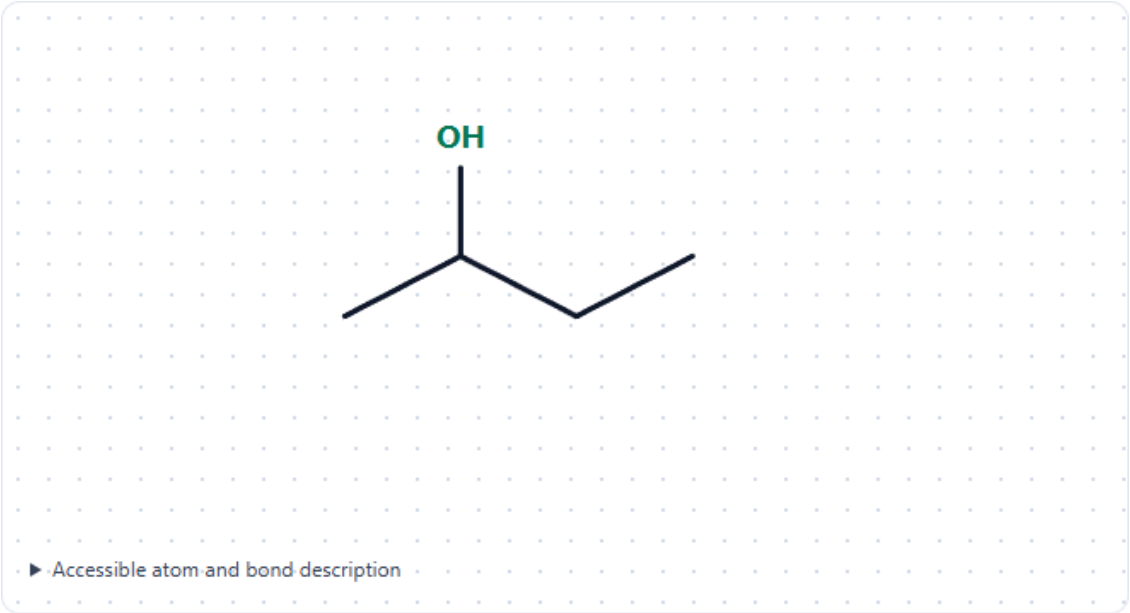
Answer type: number | unit: isomers

stereocenters:stage-3 | After accounting for the meso form, how many distinct stereoisomers exist?

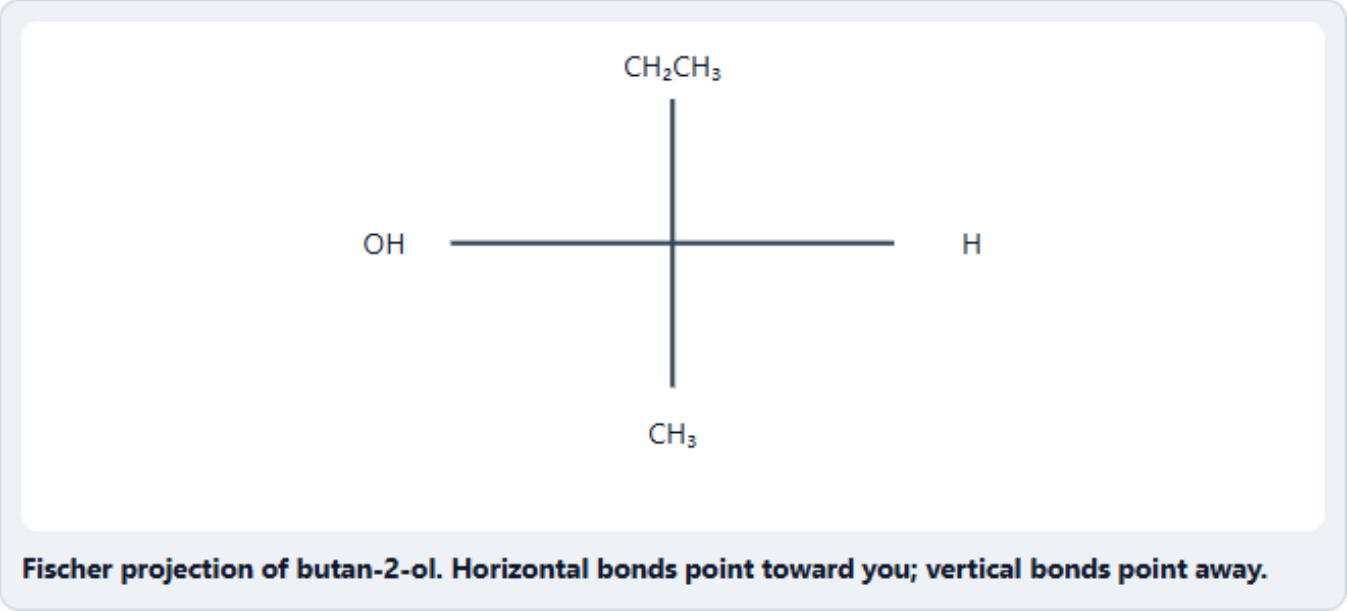
Answer type: number | unit: isomers

fischer-cip | Independent exercise

Screenshot: fischer-cip-stage-1-molecule



Screenshot: fischer-cip-diagram



{"diagramSpecification": {"type": "fischer", "top": "CH□CH□", "bottom": "CH□", "left": "OH", "right": "H", "caption": "Fischer projection of butan-2-ol. Horizontal bonds point toward you; vertical bonds point away."}}

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

4 (E): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [1, 4, 1]]

fischer-cip:stage-1 | Which substituent has highest CIP priority?

Answer type: choice

Option 0 (A): OH

Option 1 (B): Ethyl

Option 2 (C): Methyl

Option 3 (D): H

fischer-cip:stage-2 | In this projection, the lowest-priority H points…

Answer type: choice

Option 0 (A): Away

Option 1 (B): Toward you

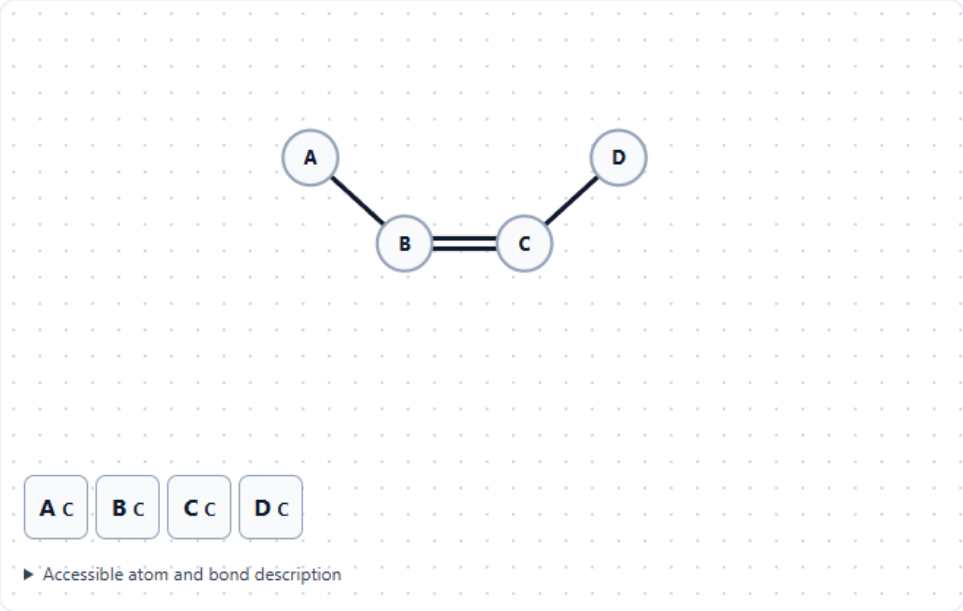
Option 2 (C): In the paper plane

fischer-cip:stage-3 | Assign the configuration: R or S.

Answer type: text

ez-ee | Independent exercise

Screenshot: ez-ee-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=60.00, y=50.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=100.00; charge=0; explicit H override=none

2 (C): C; x=185.00, y=100.00; charge=0; explicit H override=none

3 (D): C; x=240.00, y=50.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [2, 3, 1]]

ez-ee:stage-1 | Select both alkene carbon atoms.

Answer type: atoms

ez-ee:stage-2 | Assign E or Z.

Answer type: text

ez-ee:stage-3 | Separately, what is the ee of a 75:25 enantiomer mixture?

Answer type: number | unit: %